HACKATHON PROJECT PROPOSAL

BrandOS

A living brand intelligence system

A connected brand digital twin that tracks facts, rules, assets, and content - then detects outdated material and gradual brand drift before it reaches the public.

THE PROPOSAL	THE TEAM
We will build a versioned source of truth for brand knowledge, connect content to the rules and facts it relies on, and surface change impact and recurring drift with evidence for human review.	**Samriddha Chaudhury** Team Lead · Full Stack Developer **Md Rizwaan Rahaman** Full Stack Developer

SUBMISSION DETAILS

Hackathon	[Hackathon name]
Institution / team affiliation	[Add institution or affiliation]
Participant IDs	[Add required registration / participant IDs]
Contact email and phone	[Add team contact details]

Working product name: BrandOS. Confirm name availability before public use.

01 Understanding of the problem statement

Organisations publish content through employees, agencies, and AI tools. Their approved brand voice, product facts, terminology, visual assets, and claim policies are often scattered across documents and teams. This makes it easy for outdated facts, unsupported claims, inconsistent language, or obsolete assets to pass through review. The issue compounds over time: a content item may have been correct when approved, then become inaccurate after a product fact or policy changes.

We understand the challenge as both a content-review problem and a change-management problem. Teams need to check individual content against trusted sources, understand why an issue was raised, and see which existing materials may be affected when approved knowledge changes. The system must support human judgment, preserve review history, and treat uncertain findings as items for review rather than presenting guesses as facts.

02 Users affected by the problem

User group	Need
Brand and marketing teams	Keep campaigns, messaging, and assets consistent across channels and teams.
Content creators and designers	Receive clear, actionable guidance while preserving the intent of their work.
Brand managers and reviewers	Prioritise review, approve exceptions, and trace decisions to sources.
Product and communications teams	Update claims and materials when specifications or approved messages change.
Growing organisations and distributed teams	Coordinate freelancers, agencies, and regional teams without relying on one person’s memory.

03 Proposed solution

BrandOS is a living brand intelligence system. It maintains a versioned model of approved brand policies, facts, terminology, assets, campaigns, and content. During review, it links detected issues to the specific sources that support them. When a fact or policy changes, it identifies connected content that may need re-evaluation. Across review history, it highlights recurring issues and shifts that may indicate brand drift.

The product will distinguish a confirmed contradiction from a claim that simply lacks approved evidence. It will show examples and source references, explain uncertainty, and route decisions to people. It will not silently rewrite guidelines or treat an automated score as publication approval.

Signature capability

Change Impact + Drift Radar: update an approved fact or rule, see the content linked to the old version, and inspect recurring review patterns such as rising use of unapproved claims or terminology. Each signal includes examples and remains a prompt for human investigation.

04 Complete product workflow

Stage	What happens
1. Set up a brand	Create a workspace and add approved rules, facts, terminology, and assets. For the prototype, provide a prepared demo brand and allow manual entry of structured information.
2. Review and approve knowledge	A brand administrator checks proposed or edited rules and facts, marks them approved, and assigns a version. Only approved sources are used as authoritative evidence.
3. Submit content	A creator enters text and selects its channel or content type. The MVP focuses on text; visual asset review can follow if time permits.
4. Analyse with evidence	The system runs deterministic terminology and fact checks, retrieves relevant approved sources, and uses an AI model to identify semantic issues and propose explanations.
5. Resolve and route	The creator reviews findings, edits content, or requests an exception. A manager can comment, request revisions, or approve according to the demo workflow.
6. Track versions and dependencies	The system stores submissions, findings, decisions, source versions, and links between content and the facts or rules used in its review.
7. Simulate a change	An administrator updates a fact or policy. BrandOS compares connected content with the new version and produces a review queue of potentially affected items.
8. Inspect drift	A dashboard aggregates findings, overrides, and recurring phrases over time. It shows representative examples and prompts administrators to review possible trends.

05 Main features planned

Feature	Planned behaviour
Versioned brand knowledge	Structured policies, approved facts, preferred and restricted terms, source references, and effective versions.
Evidence-linked content review	Findings with flagged text, category, severity, explanation, source, and a suggested correction or review action.
Content-to-source relationships	Links showing which content references which facts, policies, and approved assets.
Change impact analysis	A fact or policy update surfaces connected content that should be rechecked, including older reviewed items where available.
Human review workflow	Creator, reviewer, and administrator actions; comments; statuses; exceptions with justification; and decision history.
Brand drift dashboard	Trends in repeated findings, overrides, terminology, and claim patterns with examples and time context.
Privacy-aware workspace	Workspace-scoped access, private file storage if files are used, server-side authorization, and clear disclosure of external AI processing.

06 Technology stack

Layer	Technology	Purpose
Web application	Next.js + TypeScript	Review interface, knowledge setup, impact view, and dashboard.
UI	Tailwind CSS + shadcn/ui	Consistent components and responsive screens.
Backend	Next.js Route Handlers	Authorization, review orchestration, versioning, and impact analysis.
Database / Auth / Storage	Supabase: PostgreSQL, Supabase Auth, private Storage	Store workspace data, user roles, review history, and private uploads.
AI	Gemini API or an available equivalent	Assist with rule extraction and semantic review; return structured, evidence-linked results.
Search	PostgreSQL full-text search	Retrieve relevant approved sources for the MVP; add vector search only if needed.
Charts / Hosting	Recharts; Vercel + Supabase	Visualise trends and deploy the demo.

Reliability approach

Keep deterministic checks for exact terminology and structured facts. Use AI to assist with interpretation and explanations, not as the sole authority. Validate each returned source against the current workspace and approved version. If no support is found, label the claim “not supported by the available approved sources” and route it for review.

07 Team responsibilities

Member	Role	Responsibilities
Samriddha Chaudhury	Team Lead · Full Stack Developer	Own product scope and integration; design the data model and backend workflow; implement brand knowledge, versioning, change-impact logic, and coordinate the end-to-end demo.
Md Rizwaan Rahaman	Full Stack Developer	Build the frontend experience for brand setup, content submission, findings, review queue, and drift dashboard; integrate review APIs and prepare representative demo content.

Shared contribution

Both members will review the user journey, connect frontend and backend, validate the demo against prepared cases, refine the pitch, and support the final presentation. If the team or event provides additional contributors, responsibilities should be updated with their names and assigned work.

08 Development plan across all three rounds

Round	Planned work and outcome
Round 1 - Proposal submission	Define the user problem, product concept, workflow, differentiator, stack, and team responsibilities. Submit this proposal PDF. Confirm event-specific requirements, participant details, and judging criteria.
Round 2 - Working prototype	Build the prepared brand workspace, structured facts and policies, text review with source-linked findings, review status flow, and a basic change-impact simulation. Seed realistic demo data. Focus on one complete scenario that works reliably.
Round 3 - Final demonstration	Polish the interface and narrative; demonstrate a fact or policy update surfacing affected content; show a recurring drift pattern backed by saved examples; explain uncertainty and human approval. Validate key scenarios, prepare the presentation, and report measured results only.

The event’s exact round dates, durations, and technical requirements were not provided. This plan follows the three-stage structure implied by a proposal round, build round, and final round; adjust it to the official schedule.

09 Expected impact

BrandOS is intended to help teams catch mismatches earlier, reduce repeated manual searching, and understand how one approved change can affect content already in circulation or under review. The drift view can help brand teams decide where clearer guidance, updated templates, or targeted training may be useful.

The prototype will demonstrate impact through traceable examples rather than unsupported outcome claims. We plan to show whether the system can identify prepared outdated facts, connect findings to the correct sources, surface affected content after a change, and reveal repeated issues from its own review history. Any performance figures will be reported only after running a defined evaluation set.

10 Future scope after the event

- Add PDF and DOCX ingestion with administrator review of extracted rules and facts.
- Add image OCR, approved-logo comparison, visual asset versioning, and measurable contrast checks.
- Expand drift analysis across channels, departments, campaigns, and longer time periods.
- Add stronger retrieval and evaluation tooling, including version-aware regression tests for policies.
- Integrate with content management and publishing systems to provide a configurable pre-publication review gate.
- Add organisation-level controls such as audit exports, retention settings, and more granular roles.
- Support multilingual brand rules and region-specific approved facts.

11 Team and participant details

The team currently has two listed participants. The fields below should be completed using the event's official registration details before submission; no contact or institution details were provided in the project brief.

Field	Samriddha Chaudhury	Md Rizwaan Rahaman
Team role	Team Lead	Full Stack Developer
Participant / registration ID	[Add ID]	[Add ID]
Institution / affiliation	[Add institution]	[Add institution]
Email	[Add email]	[Add email]
Phone / contact	[Add phone if required]	[Add phone if required]

12 Each member's role and planned contribution

Samriddha Chaudhury - Team Lead and Full Stack Developer. Leads scope, product decisions, backend architecture, and integration. Plans to implement the versioned brand data model, source relationships, change-impact workflow, and overall application integration. Shares responsibility for testing the end-to-end scenario and presenting the solution.

Md Rizwaan Rahaman - Full Stack Developer. Plans to implement the user-facing screens for brand setup, content review, findings, approval flow, and drift insights; connect those screens to backend services; and prepare the content and examples used in the demonstration. Shares responsibility for integration, validation, and presentation.